# Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau


**Team ID: LTVIP2026TMIDS89796**

Team Size: 4

**Team Leader:** Addanki Tharun Sai

Role: Operations and R&D Coordinator

**Team Member:** Adipudi Sivanagaraju

Role: Website Development.

**Team Member:** Ch Venkat Teja

Role: Tableau Dashboard Development and Reporting

**Team Member:** Goli Sri Godha

Role: Flask Development, Documentation

# 1. INTRODUCTION:

The preservation and understanding of cultural and natural heritage sites have become increasingly important in today's world, as these sites represent historical identity, environmental value, and global diversity. UNESCO World Heritage Sites are recognized locations that hold outstanding universal significance, and analyzing their distribution, categories, and risk status can provide valuable insights into global heritage conservation. However, large datasets related to heritage sites are often complex and difficult to interpret without proper visualization tools.

This project focuses on transforming UNESCO heritage site data into an interactive and visually engaging analytical platform. By using Tableau for data visualization and Flask for web integration, the project presents dashboards that allow users to explore heritage site trends across different regions, categories, and time periods. The system combines maps, charts, and analytical views to make information easier to understand and more accessible to users.

The main goal of the project is to bridge the gap between raw data and meaningful insights through modern visualization techniques. Instead of relying on static reports, the developed web application provides an interactive environment where users can analyze heritage site distributions, examine endangered locations, and observe historical patterns. This approach not only improves data understanding but also demonstrates the practical application of data analytics, dashboard development, and web technologies in a real-world scenario.

## *1.1. Project Overview:*

The UNESCO World Heritage Sites Project is an interactive data analytics and visualization system designed to analyze global heritage site information using modern visualization tools and web technologies. UNESCO World Heritage Sites represent locations of cultural, natural, and mixed importance that require protection and awareness. Understanding patterns such as regional distribution, category classification, danger status, and historical inscription trends helps researchers, students, and policymakers gain meaningful insights from large datasets.

In this project, historical UNESCO heritage site data is collected and analyzed using Tableau, where multiple visualizations such as maps, treemaps, bubble charts, category comparisons, and time-series analysis are created. These individual visualizations are then combined into structured dashboards and story points to provide a complete analytical narrative. To make the dashboards accessible through a web interface, a Flask-based web application is developed, which embeds Tableau Public visualizations using iframes. The web interface presents dashboards in a structured layout with clear navigation, allowing users to interact with data directly.

The project focuses on transforming raw heritage site data into visually appealing insights, enabling users to explore patterns such as the number of sites per region, endangered heritage locations, category distribution, and historical trends. By integrating data analytics with web deployment, the project demonstrates a practical implementation of data visualization in a real-world application environment.

### *1.2. Purpose:*

The primary purpose of this project is to provide an interactive and user-friendly platform for analyzing UNESCO World Heritage Sites through data visualization. Instead of presenting information in static tables, the system converts complex datasets into meaningful graphical representations that improve understanding and decision-making. The project aims to highlight global heritage trends, identify regions with a higher number of endangered sites, and provide insights into how heritage site inscriptions have evolved over time.

Another important objective is to demonstrate technical integration between data analytics tools and web technologies. By combining Tableau dashboards with a Flask web application, the project showcases how analytical content can be deployed as a functional website. This helps users learn practical concepts such as embedding dashboards, designing responsive layouts, and building a data-driven web interface.

Furthermore, the project supports educational awareness by encouraging users to explore heritage preservation issues through visual storytelling. It allows viewers to interact with dashboards, observe data patterns, and gain a deeper understanding of the importance of cultural and natural heritage sites worldwide. Overall, the project serves both as an analytical study and as a technical implementation that bridges data visualization with modern web development practices.

## 2. IDEATION PHASE:

The ideation phase plays a crucial role in transforming an initial idea into a structured project plan. In this stage, different concepts, user needs, and possible solutions were carefully analyzed to design an effective system for visualizing UNESCO World Heritage Site data. The main objective of the ideation phase was to identify existing challenges in understanding heritage datasets and to develop a solution that combines data analytics with interactive web technology.

During this phase, the focus was placed on understanding how users interact with large datasets and what type of visual representation would make information easier to interpret. Various approaches were explored, including static reports, standalone dashboards, and fully interactive web applications. After evaluating these options, the idea of creating an integrated analytics platform using Tableau dashboards embedded within a Flask-based website was selected. This approach ensures that users can explore heritage site information through maps, charts, and storytelling features in a single platform.

The ideation phase also involved analyzing user expectations through empathy mapping, identifying key analytical features such as regional distribution, category comparison, endangered site analysis, and historical trend forecasting. Brainstorming sessions helped finalize the project structure, decide on visualization types, and plan the layout of dashboards and web pages. By the end of this phase, a clear roadmap was established, defining how data would be processed, visualized, and presented to users in an engaging and informative manner. Overall, the ideation phase laid the foundation for developing a user-centered and data-driven analytics system that highlights the importance of UNESCO World Heritage Sites through modern visualization techniques.

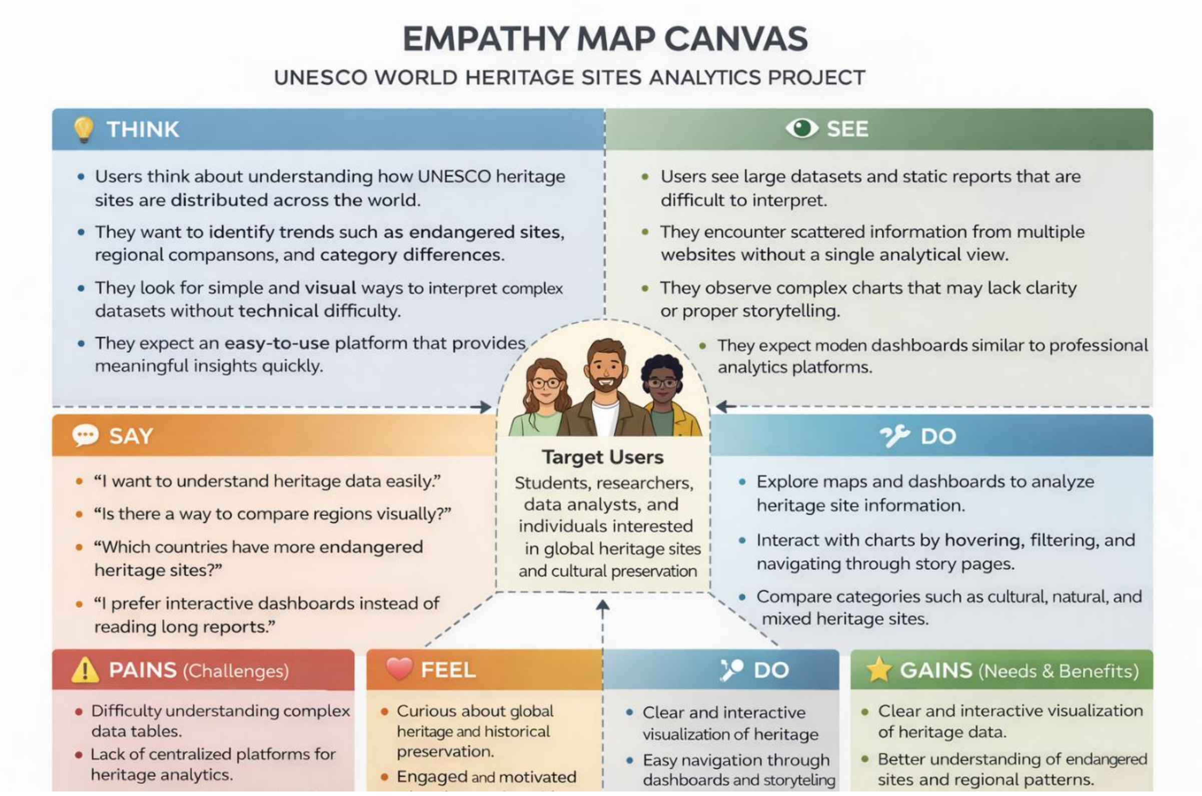
## *2.1. Problem Statement:*

UNESCO World Heritage Sites hold significant cultural, historical, and environmental value, yet the data related to these sites is often available only in complex datasets or static reports. Such formats make it difficult for users to explore patterns, identify trends, or understand the overall distribution of heritage locations worldwide. Many users face challenges in analyzing which regions have more heritage sites, how categories differ, or how the number of endangered sites changes over time. Without interactive tools, extracting meaningful insights from raw data becomes time-consuming and less effective.

Another major challenge is the lack of a unified platform that combines data analysis with an accessible web interface. Traditional analysis methods may present information in tables or text-heavy documents, which reduces user engagement and understanding. Therefore, the problem addressed in this project is the need to design an interactive and visually driven analytics system that simplifies UNESCO heritage data through dashboards, maps, and storytelling features. The solution aims to make heritage site information easier to interpret while also demonstrating modern data visualization and web integration practices.

## *2.2. Empathy Map Canvas:*

The Empathy Map Canvas was used to better understand the target audience and guide the design of the project. The primary users include students, researchers, data analysts, and individuals interested in global heritage conservation. These users often think about exploring heritage trends and learning more about cultural and natural sites, but they may lack advanced analytical tools. They see scattered information from different online resources, making it difficult to obtain a clear overall picture. They hear discussions about heritage preservation and global awareness but may struggle to connect this information with real data insights.




From the user perspective, they say and do tasks such as searching for heritage site details, comparing countries, or exploring maps to understand site locations. At the same time, they feel confused or overwhelmed when presented with raw datasets without proper visualization. By analyzing these behaviors and emotions, the project focuses on creating an intuitive dashboard interface that supports exploration, simplifies complex information, and enhances user engagement. The empathy mapping process ensured that the final system addresses real user needs by emphasizing clarity, interactivity, and ease of use.

## *2.3. Brainstorming:*

The brainstorming phase involved generating and evaluating multiple ideas to determine the most effective way to present UNESCO heritage data. Initially, several approaches were considered, including building a static analytical report, creating standalone dashboards, or developing a full web-based analytics platform. After discussion and evaluation, the idea of combining Tableau dashboards with a Flask web application was selected, as it provides both powerful visualization capabilities and an interactive user interface.

Different visualization concepts were explored during brainstorming, such as using maps to represent geographical distribution, pie charts for category comparison, treemaps for highlighting top endangered sites, bubble charts for regional analysis, and forecasting models to show future trends. The team also discussed layout design, user navigation, and how storytelling features could guide viewers through the analysis step by step. This phase helped identify the key objectives of the project, define the structure of dashboards, and ensure that the final design would be informative, visually appealing, and easy to understand. The brainstorming process ultimately shaped the project into a comprehensive analytics solution that balances technical implementation with user-centered design.

# 3. REQUIREMENT ANALYSIS:

Requirement analysis is a crucial phase in the development of the UNESCO World Heritage Sites analytics project, as it defines the functional expectations, user needs, and technical resources required to build a successful system. This phase ensures that the project aligns with real-world requirements by identifying how users will interact with the application, what type of insights they expect, and how the system should be structured to provide a smooth analytical experience. By clearly outlining requirements, the project minimizes development challenges and establishes a strong foundation for design and implementation.

The requirement analysis process involved understanding the target users, defining system objectives, and identifying the tools necessary to transform raw heritage data into interactive visual insights. The system is designed to support users who want to explore UNESCO heritage information in a structured and visually engaging way. Key considerations included creating intuitive navigation, ensuring responsive dashboard embedding, and providing meaningful analytical features such as category comparison, regional distribution analysis, endangered site tracking, and historical trend visualization.

## *3.1. Customer Journey Map:*

The Customer Journey Map explains the complete user interaction process with the system. The journey begins when a user accesses the web application through a browser and lands on the homepage. At this stage, users are introduced to the project through a header section and organized dashboard panels. They then navigate through the embedded Tableau dashboards to explore heritage site insights. Users may start by viewing an overview dashboard that presents global distribution and category analysis, followed by a deeper exploration of endangered site trends through the second dashboard.

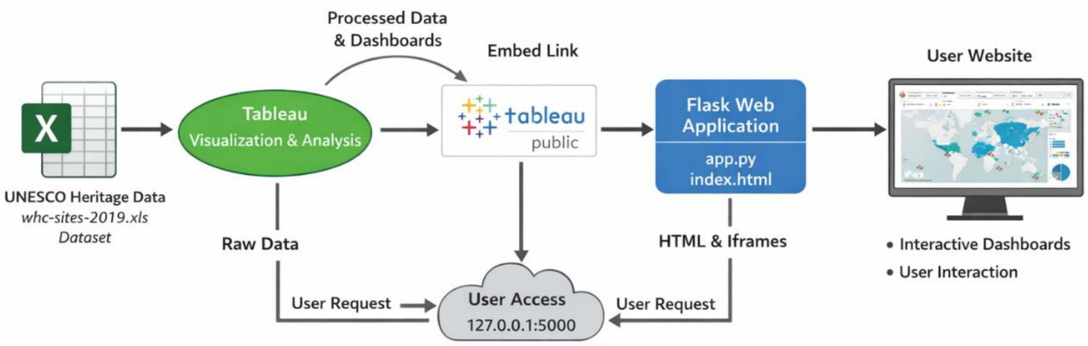
As users interact with the visualizations, they can hover over charts, explore maps, and follow story points that guide them through the analysis step by step. The interface is designed to minimize complexity and provide a smooth experience by presenting information in structured sections. The journey concludes when users gain a clear understanding of heritage site patterns, enabling them to interpret data effectively without needing advanced technical knowledge.

## *3.2. Solution Requirement:*

The solution requirements describe both functional and non-functional aspects of the system. From a functional perspective, the project must support interactive visualization, web integration, and structured storytelling. The system should display multiple dashboards, allow smooth navigation between sections, and present heritage site information through visually appealing charts. The ability to embed Tableau Public dashboards within a Flask web application is a key requirement, ensuring that users can interact with live analytics directly from the website.

Non-functional requirements focus on performance, usability, scalability, and reliability. The application should load dashboards efficiently without affecting responsiveness. The interface must be simple and intuitive, allowing users to understand data without confusion. The design should also be flexible so that additional dashboards or datasets can be integrated in the future. Security and stability considerations include maintaining proper routing in Flask and ensuring that external embedded content loads safely within the webpage.

## *3.3. Data Flow Diagram:*




## *3.4. Technology Stack:*

The technology stack used in this project combines analytical tools and web development frameworks to create a complete data-driven system. Tableau is the core visualization platform used to design interactive dashboards, create maps, perform trend analysis, and develop story-based presentations. Tableau Public serves as the hosting environment for published dashboards, enabling easy sharing and embedding through web links.

The backend of the web application is built using Flask, a lightweight Python framework that manages server-side routing and loads the main webpage. HTML is used to structure the content of the website, including headers, sections, and embedded dashboards, while CSS is used to style the interface with a modern and visually appealing design. This combination of technologies ensures seamless integration between analytics and web deployment, allowing the project to function as a fully interactive heritage site analytics platform.

# 4. PROJECT DESIGN:

Project design is an essential phase that transforms identified requirements into a structured system plan. In this stage, the overall layout, architecture, and workflow of the UNESCO World Heritage Sites analytics platform are carefully planned to ensure that the final solution meets user needs and technical objectives. The design focuses on integrating data visualization with a web-based interface so that users can explore heritage site insights through an interactive and visually organized environment.

The project design also emphasizes usability, scalability, and clarity. By defining how Tableau dashboards connect with the Flask web application, the system ensures smooth navigation and efficient data presentation. This phase outlines how each component data source, visualization layer, web framework, and user interface works together to create a unified analytics platform that supports storytelling and interactive exploration.

## *4.1. Problem Solution Fit:*

The problem solution fit explains how the proposed system effectively addresses the challenges identified during earlier phases. Users often struggle to interpret large heritage datasets because traditional data formats lack visual clarity and interactivity. The project design solves this issue by converting complex data into intuitive visual dashboards, allowing users to explore information through maps, charts, and analytical views rather than static tables. This approach improves data accessibility and enhances understanding of global heritage patterns.

Additionally, the system aligns with user expectations by providing an easy-to-use web interface that requires no advanced technical skills. Instead of building a complex standalone application, the design leverages Tableau’s powerful visualization features and integrates them into a Flask-based website. This ensures that the solution is practical, scalable, and capable of delivering meaningful insights while maintaining a clean and structured user experience.

## *4.2. Proposed Solution:*

The proposed solution is an interactive analytics platform that combines Tableau dashboards with a Flask web application to present UNESCO heritage site data in a visually engaging format. Tableau is used to create multiple dashboards that analyze regional distribution, site categories, endangered locations, and historical trends. These dashboards are published on Tableau Public and embedded into the website using iframe integration, enabling users to interact with live visualizations directly within the web interface.

The solution also focuses on creating a clear navigation structure that organizes dashboards into sections such as overview analysis, danger site evaluation, and storytelling pages. By integrating frontend design with backend routing, the project provides a seamless browsing experience where users can move between dashboards without confusion. This proposed approach ensures that analytical insights are delivered in a structured and professional manner while maintaining flexibility for future enhancements.

### *4.3. Solution Architecture:*

The solution architecture describes how different components of the project interact to deliver the final output. At the base level, the UNESCO heritage dataset acts as the primary data source. This data is processed and visualized in Tableau, where analytical dashboards and story points are created. Once published on Tableau Public, the dashboards generate embed links that are integrated into the Flask web application. Flask manages routing and loads HTML pages that display the dashboards through embedded iframes.

From a system perspective, the architecture follows a layered structure consisting of the data layer, visualization layer, application layer, and presentation layer. The data layer includes the Excel dataset, the visualization layer consists of Tableau dashboards, the application layer is handled by Flask, and the presentation layer is built using HTML and CSS. This architecture ensures efficient communication between components while maintaining modularity, making it easier to update visualizations or improve the web interface without affecting the entire system.

## 5. PROJECT PLANNING & SCHEDULING

Project planning and scheduling are essential steps that ensure the successful development and completion of the UNESCO World Heritage Sites analytics project. This phase focuses on organizing tasks, defining timelines, and allocating resources effectively to achieve project goals within a structured framework. Proper planning helps in reducing risks, maintaining consistency, and ensuring that each stage of development — from data analysis to web deployment — progresses smoothly.

The project was divided into multiple phases such as ideation, requirement analysis, design, development, testing, and deployment. Each phase was carefully planned to ensure that specific objectives were met before moving to the next stage. Scheduling also played an important role in balancing analytical work in Tableau with web integration using Flask. By following a step-by-step timeline, the project maintained clarity in development and allowed continuous improvements throughout the process.

### *5.1. Project Planning:*

Project planning involved identifying key tasks and organizing them into a logical sequence to build the final system efficiently. The initial stage included understanding the UNESCO heritage dataset, defining project objectives, and selecting appropriate tools such as Tableau for visualization and Flask for web development. After this, detailed planning was carried out to design dashboards, create story points, and structure the web interface. Each activity was aligned with project goals to ensure that both analytical and technical components worked together seamlessly.

Another important aspect of planning was risk management and flexibility. Potential challenges such as dashboard layout issues, embedding errors, or performance concerns were considered during planning to avoid delays during implementation. Regular evaluation of progress helped ensure that tasks were completed on time and met quality expectations. Overall, project planning provided a clear roadmap that guided the development process from concept to deployment, ensuring that the final system remained organized, efficient, and user-focused.

## 6. FUNCTIONAL AND PERFORMANCE TESTING:

Functional and performance testing are essential stages in the development process of the UNESCO World Heritage Sites analytics project. These testing phases ensure that the system operates correctly, meets user expectations, and performs efficiently under normal usage conditions. Functional testing focuses on verifying whether each feature of the application works according to the project requirements, while performance testing evaluates how well the system handles loading, responsiveness, and user interaction with embedded dashboards.

The testing process was carried out by running the Flask application locally and interacting with the embedded Tableau dashboards through a web browser. Various scenarios such as dashboard loading, story navigation, iframe embedding, and user interaction with charts were examined to confirm that the application behaves as expected. Through these tests, the system was evaluated for reliability, usability, and overall user experience.

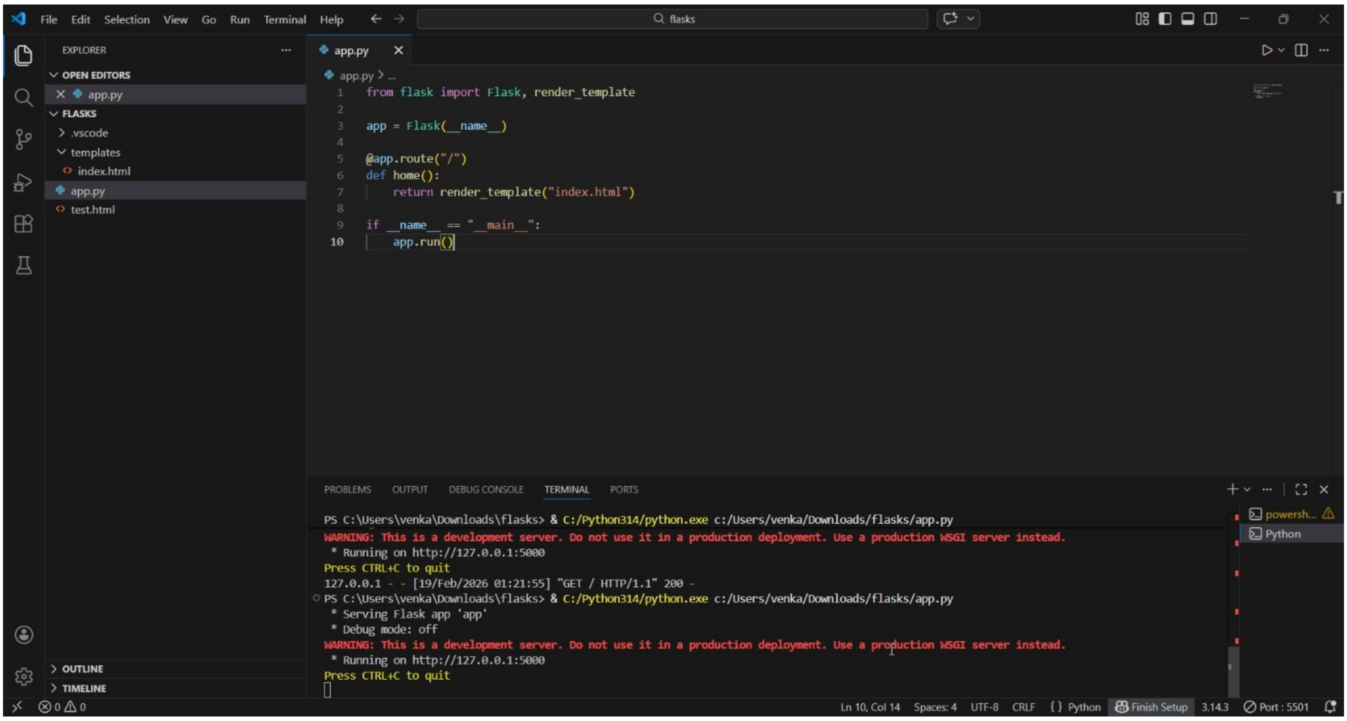
### *6.1. Performance Testing:*

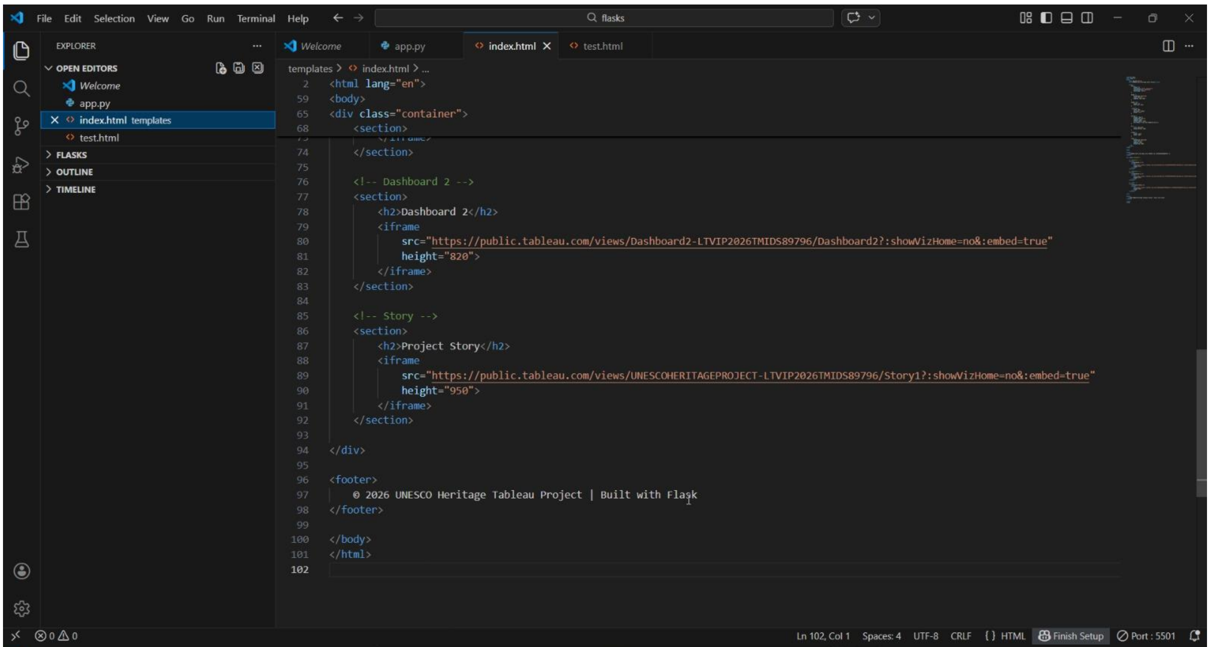
Performance testing was conducted to measure how efficiently the web application loads and displays Tableau dashboards without affecting responsiveness. Since the project uses embedded Tableau Public visualizations, it was important to ensure that the dashboards load smoothly within the webpage and maintain interactivity. The application was tested by accessing the Flask server multiple times and observing dashboard loading times, navigation speed, and responsiveness when interacting with charts such as maps, treemaps, and forecasting graphs.

The results indicated that the system performs efficiently under normal conditions, with dashboards loading successfully and user interactions remaining smooth. Minor adjustments such as optimizing iframe height, improving layout alignment, and organizing sections helped enhance performance and visual clarity. Overall, performance testing confirmed that the project delivers a stable and responsive analytics experience, allowing users to explore UNESCO heritage site data without delays or technical issues.

# 7. RESULTS:

## VS Code Results:

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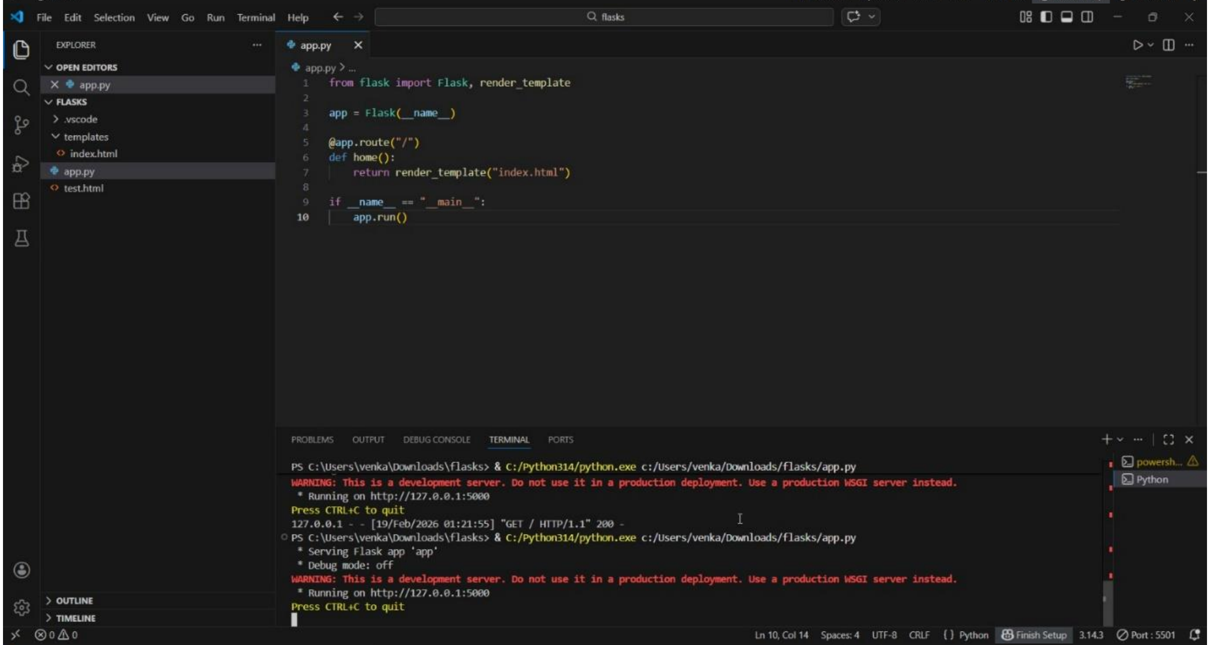
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* Running on http://127.0.0.1:5000
Press CTRL+C to quit
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PS C:\Users\venka\Downloads\flasks> & C:/Python314/python.exe c:/Users/venka/Downloads/flasks/app.py
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* Running on http://127.0.0.1:5000
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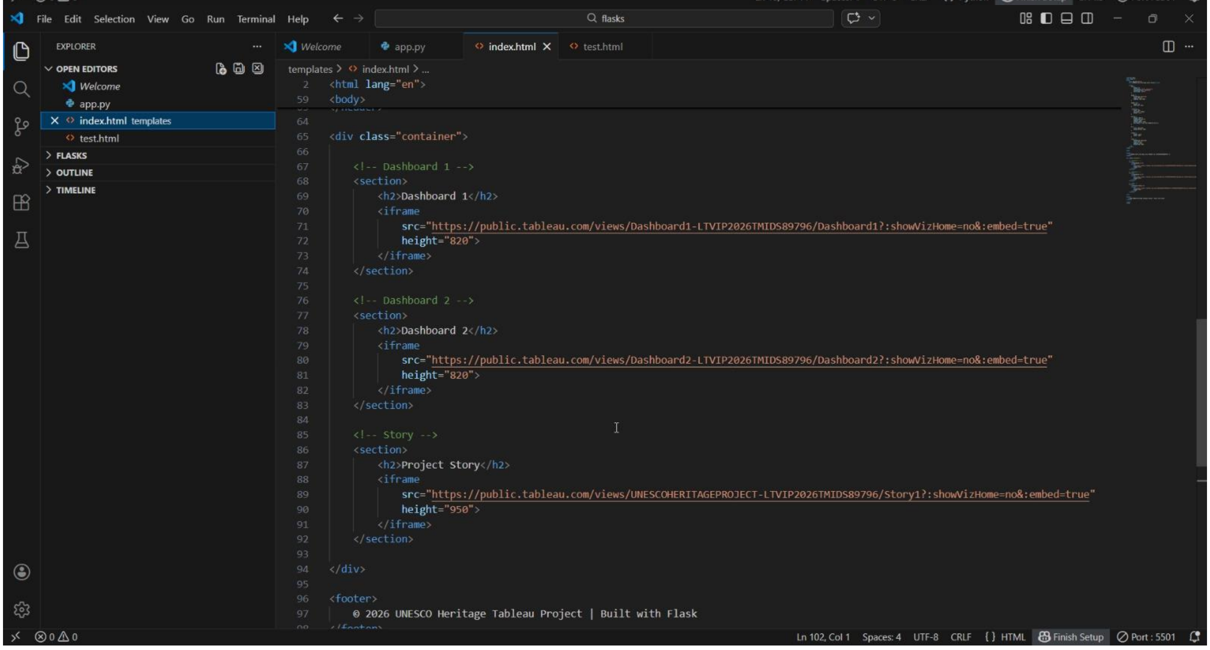
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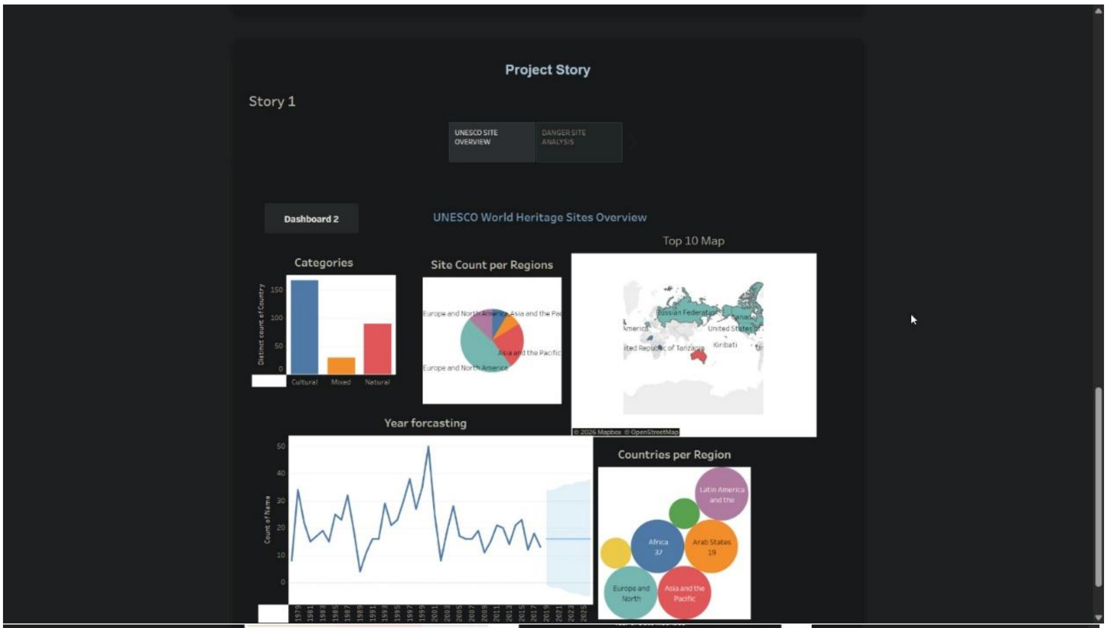
Project Story
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UNESCO SITE OVERVIEW
DANGER SITE ANALYSIS
Dashboard 2
UNESCO World Heritage Sites Overview
Top 10 Map
Categories
Site Count per Regions
Year forcasting
Countries per Region
View on Tableau Public
Share


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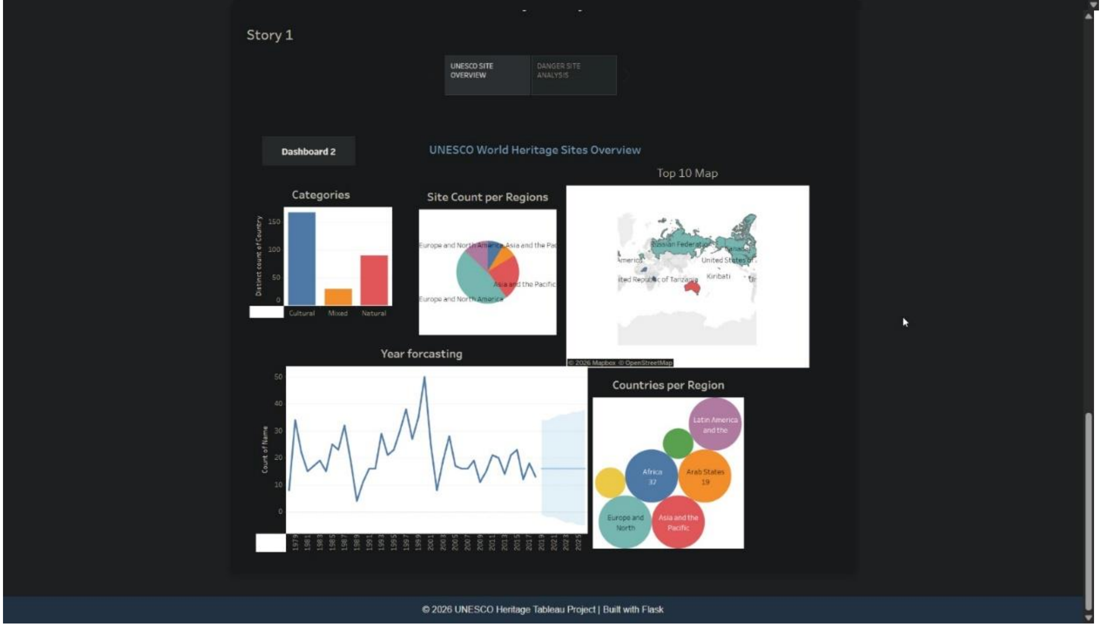
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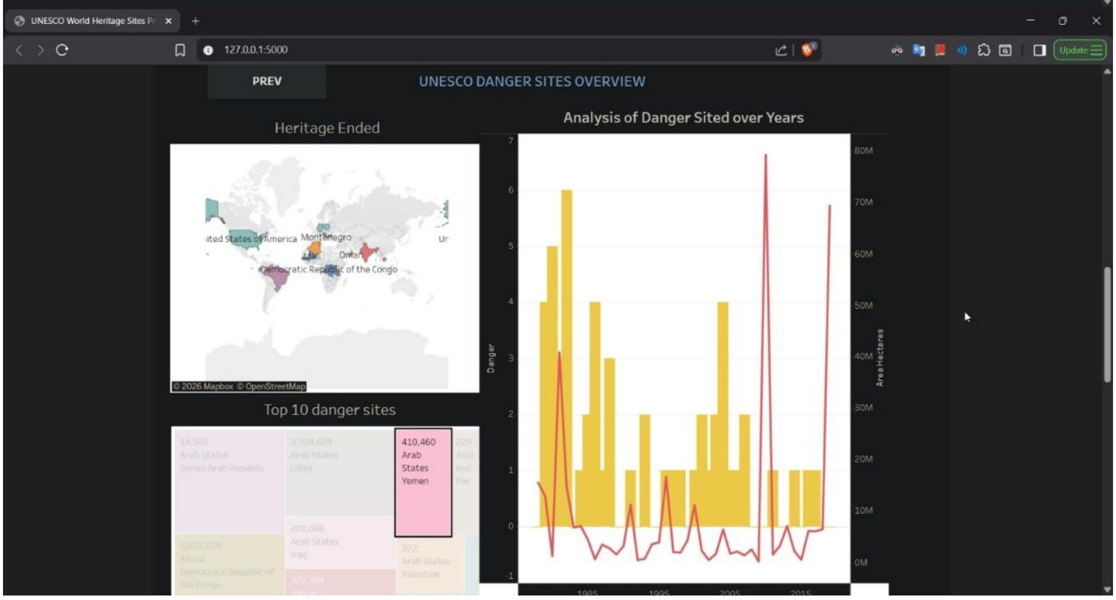
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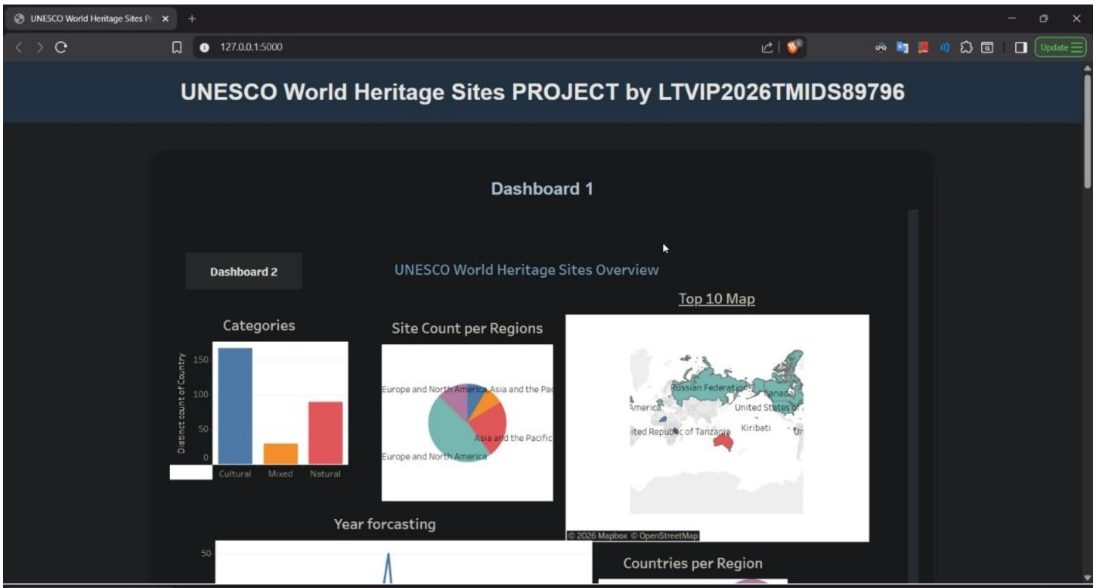
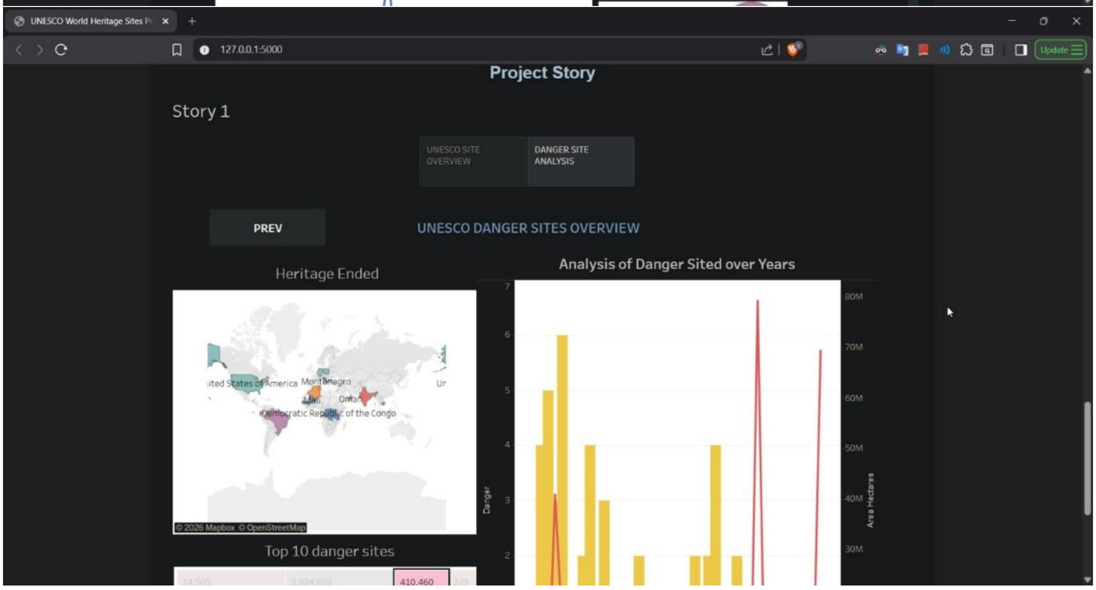
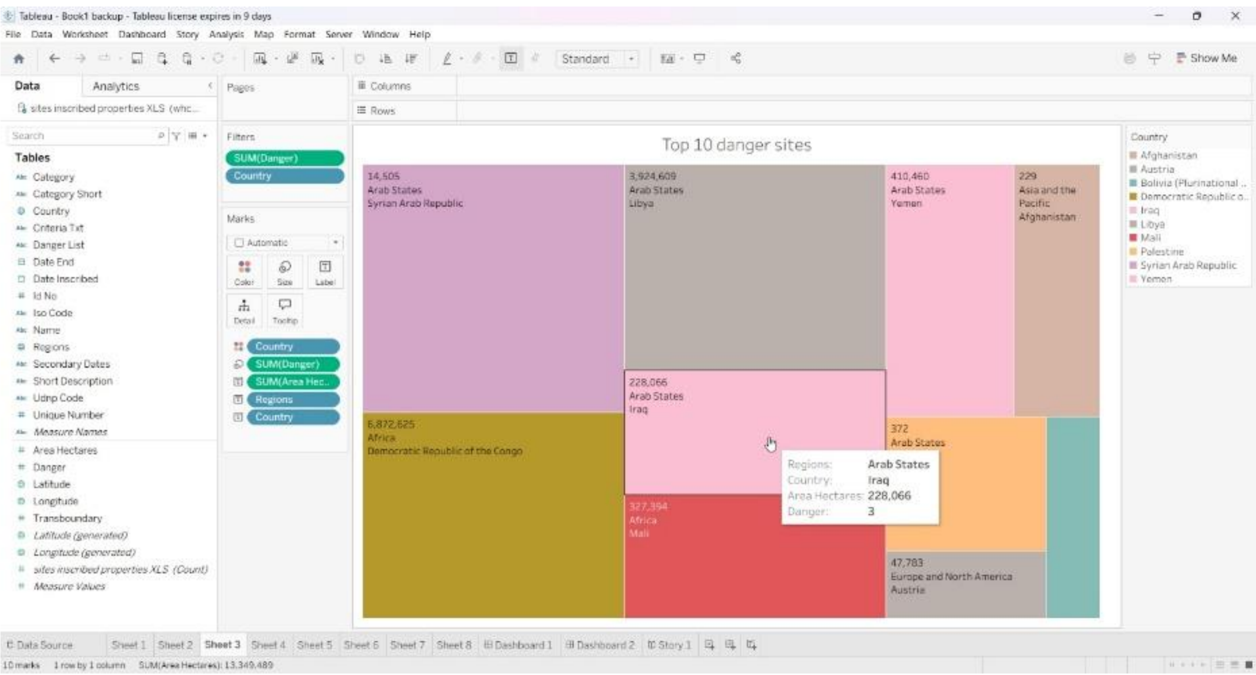
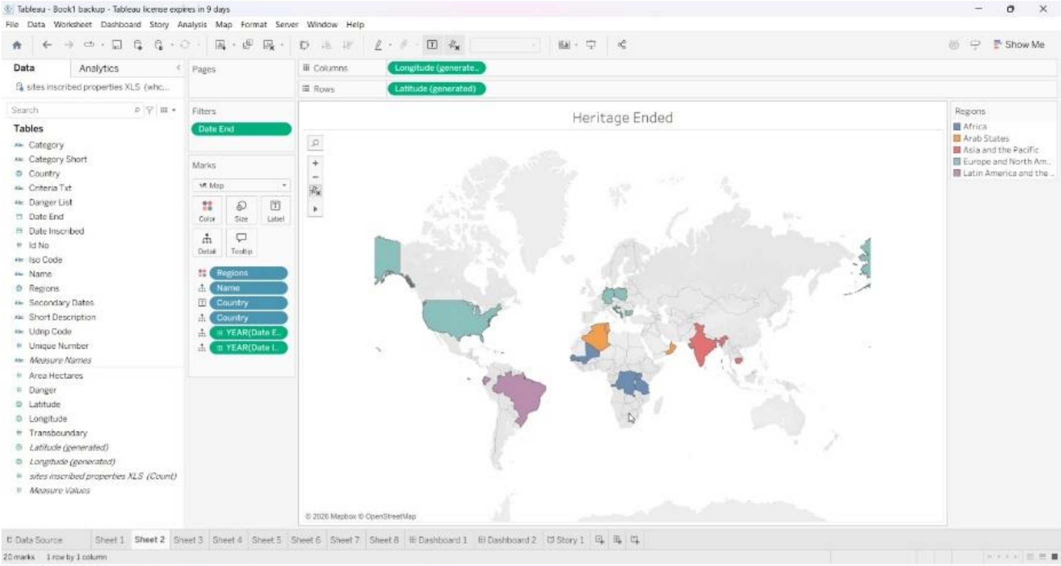



# Tableau Results:

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Dashboard 2
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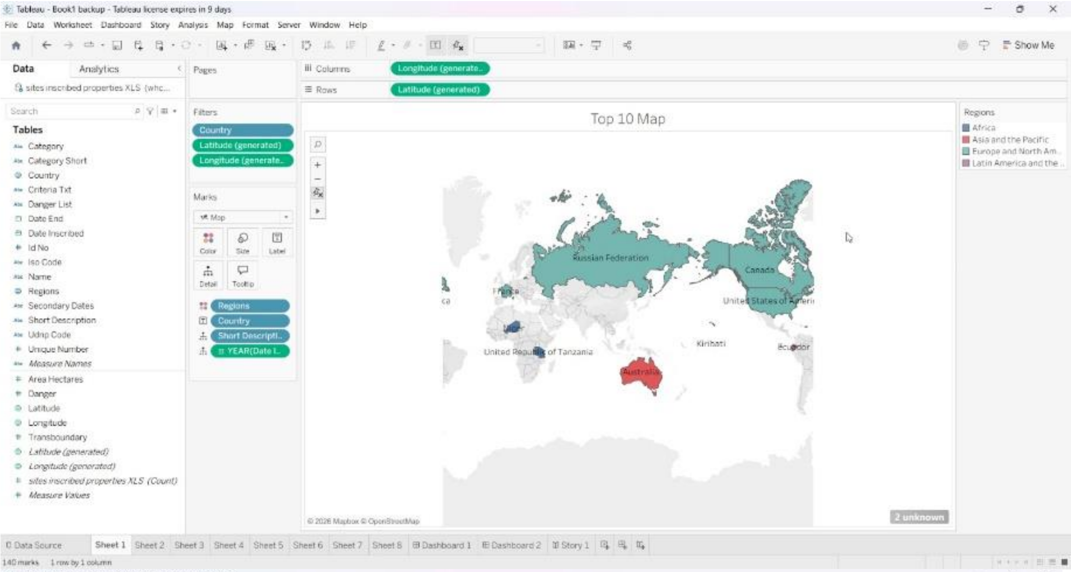
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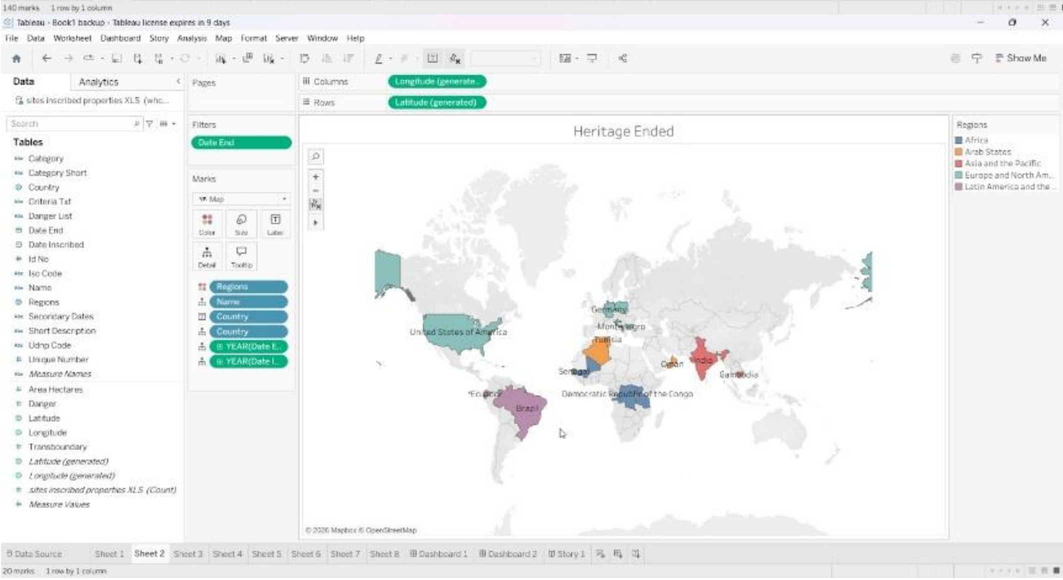
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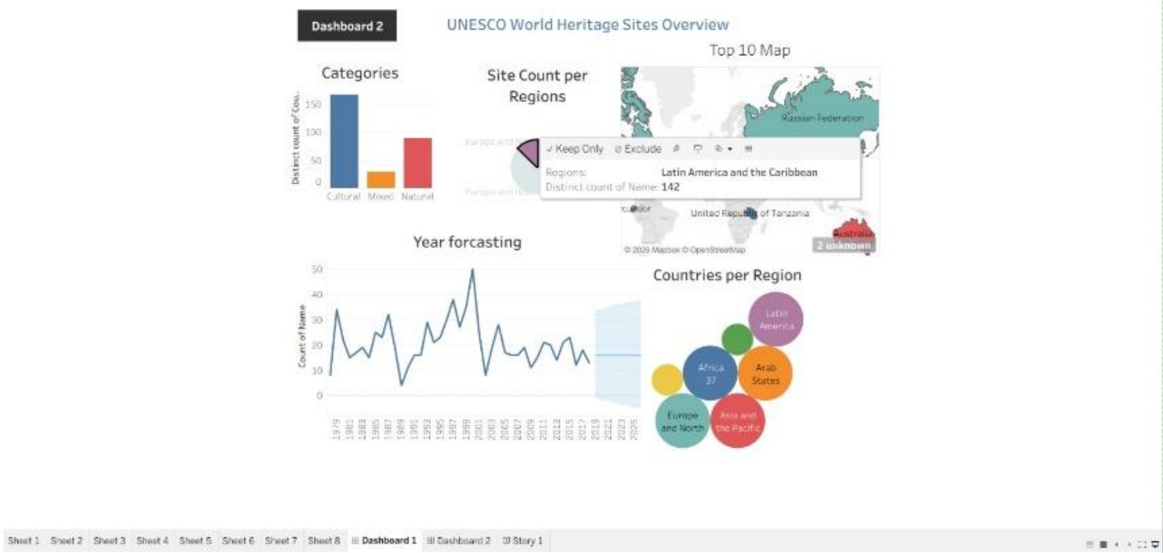
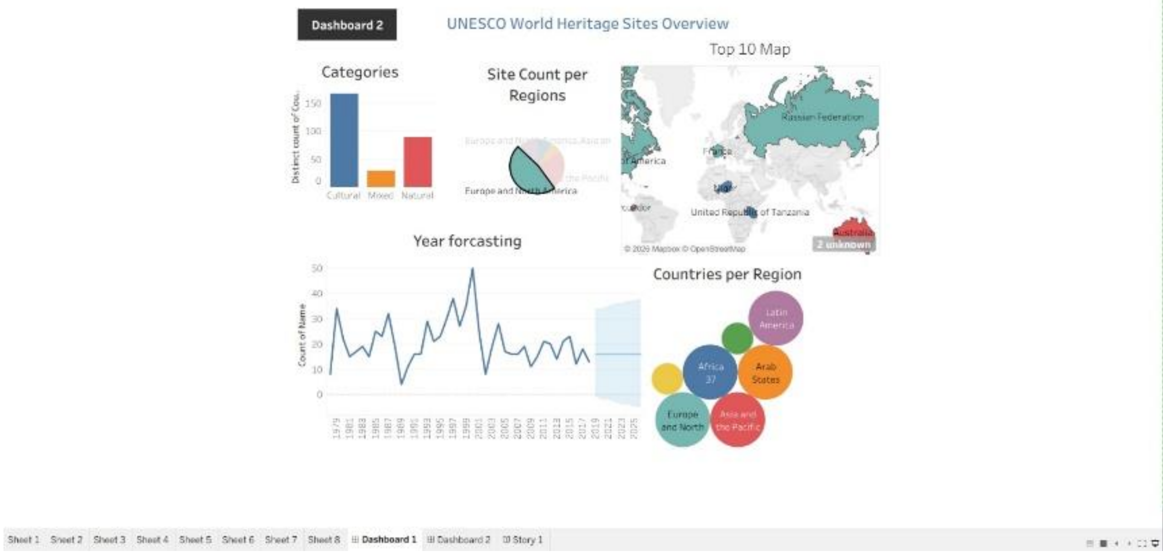
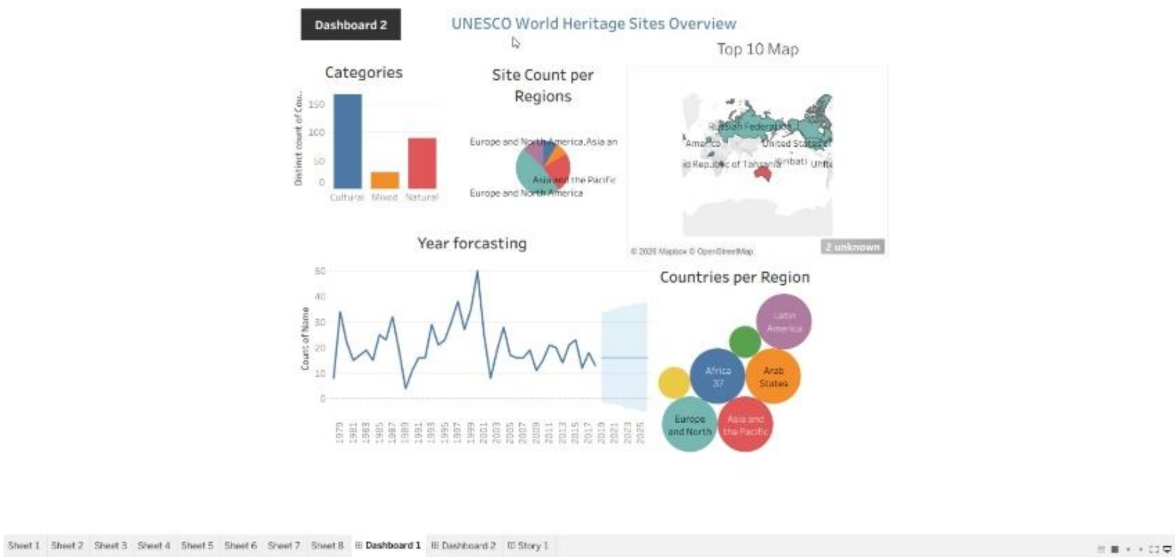
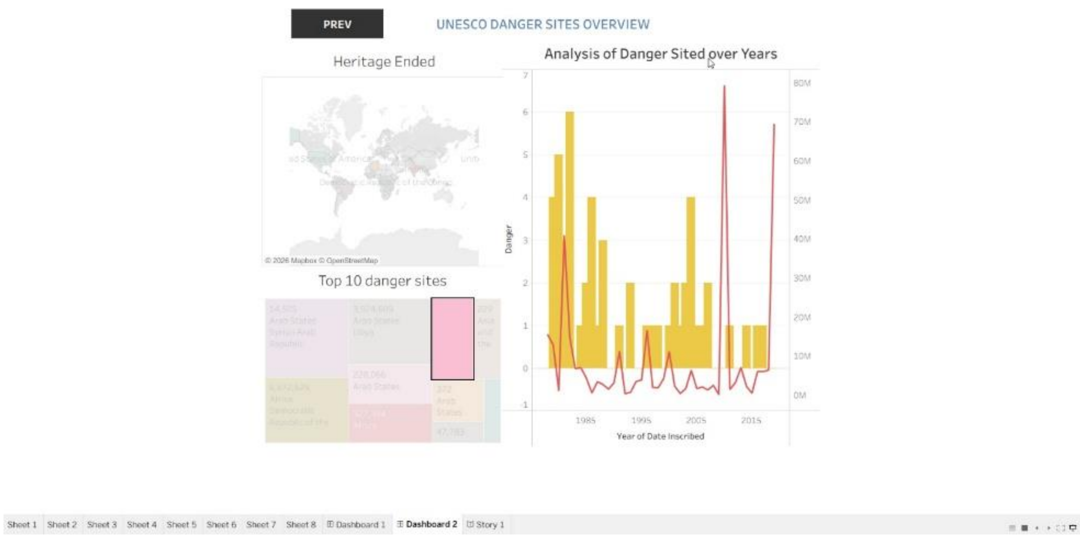
Show Me
Data
Analytics
sites inscribed properties XLS (whc...
Pages
Columns
Longitude (generate..
Rows
Latitude (generated)
Filters
Date End
Marks
Map
Regions
Name
Country
Country
YEAR(Date E..
YEAR(Date I..
Heritage Ended
Regions
Africa
Arab States
Asia and the Pacific
Europe and North Am..
Latin America and the ..
Germany
Montenegro
Tunisia
United States of America
Senegal
Oman
India
Cambodia
Ecuador
Brazil
Democratic Republic of the Congo
© 2026 Mapbox © OpenStreetMap
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
Dashboard 2
Story 1
20 marks
1 row by 1 column


Dashboard 2
UNESCO World Heritage Sites Overview
Categories
Site Count per Regions
Top 10 Map
Keep Only
Exclude
Regions: Latin America and the Caribbean
Distinct count of Name: 142
Russian Federation
United Republic of Tanzania
Australia
2 unknown
Cultural
Mixed
Natural
Year forcasting
Countries per Region
Latin America
Africa 37
Arab States
Europe and North
Asia and the Pacific
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
Dashboard 2
Story 1


Dashboard 2
UNESCO World Heritage Sites Overview
Categories
Site Count per Regions
Top 10 Map
Europe and North America
Russian Federation
France
United Republic of Tanzania
Australia
2 unknown
Cultural
Mixed
Natural
Year forcasting
Countries per Region
Latin America
Africa 37
Arab States
Europe and North
Asia and the Pacific
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
Dashboard 2
Story 1


Dashboard 2
UNESCO World Heritage Sites Overview
Categories
Site Count per Regions
Top 10 Map
Europe and North America,Asia an
Asia and the Pacific
Europe and North America
Russian Federation
United States
2 unknown
Cultural
Mixed
Natural
Year forcasting
Countries per Region
Latin America
Africa 37
Arab States
Europe and North
Asia and the Pacific
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
Dashboard 2
Story 1


PREV
UNESCO DANGER SITES OVERVIEW
Heritage Ended
Analysis of Danger Sited over Years
Top 10 danger sites
Danger
Year of Date Inscribed
1985
1995
2005
2015
Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Dashboard 1 Dashboard 2 Story 1


Story 1
UNESCO SITE OVERVIEW
DANGER SITE ANALYSIS
Dashboard 2
UNESCO World Heritage Sites Overview
Top 10 Map
Categories
Site Count per Regions
Distinct count of Cou..
Cultural
Mixed
Natural
Europe and North America
Asia and the Pacific
2 unknown
Year forcasting
Count of Name
Countries per Region
Latin America
Africa 37
Arab States
Europe and North
Asia and the Pacific
Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Dashboard 1 Dashboard 2 Story 1

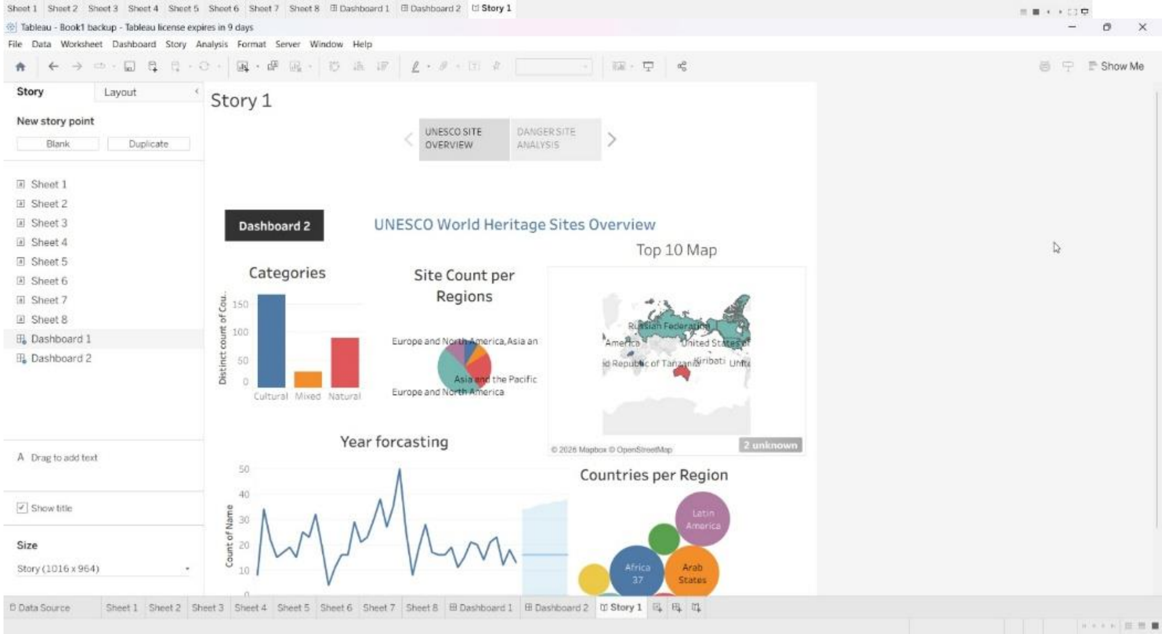
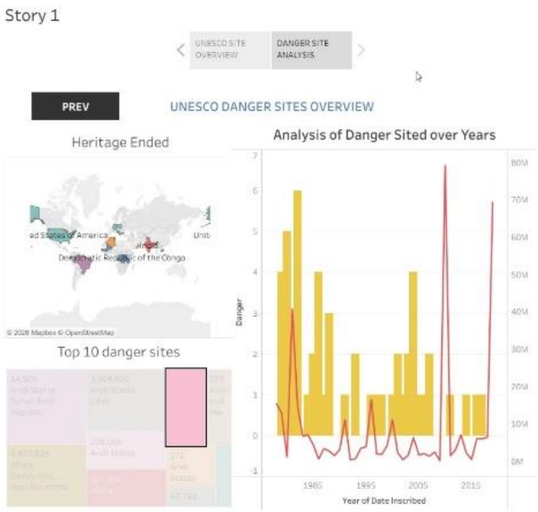


Tableau - Book1 backup - Tableau license expires in 9 days
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Show Me
Story
Layout
New story point
Blank
Duplicate
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
Dashboard 2
Drag to add text
Show title
Size
Story (1016 x 964)
Story 1
UNESCO SITE OVERVIEW
DANGER SITE ANALYSIS
Dashboard 2
UNESCO World Heritage Sites Overview
Top 10 Map
Categories
Site Count per Regions
Cultural
Mixed
Natural
2 unknown
Year forcasting
Countries per Region
Latin America
Africa 37
Arab States
Data Source Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Dashboard 1 Dashboard 2 Story 1


Story 1
UNESCO SITE OVERVIEW
DANGER SITE ANALYSIS
PREV
UNESCO DANGER SITES OVERVIEW
Heritage Ended
Analysis of Danger Sited over Years
Top 10 danger sites
Danger
Year of Date Inscribed

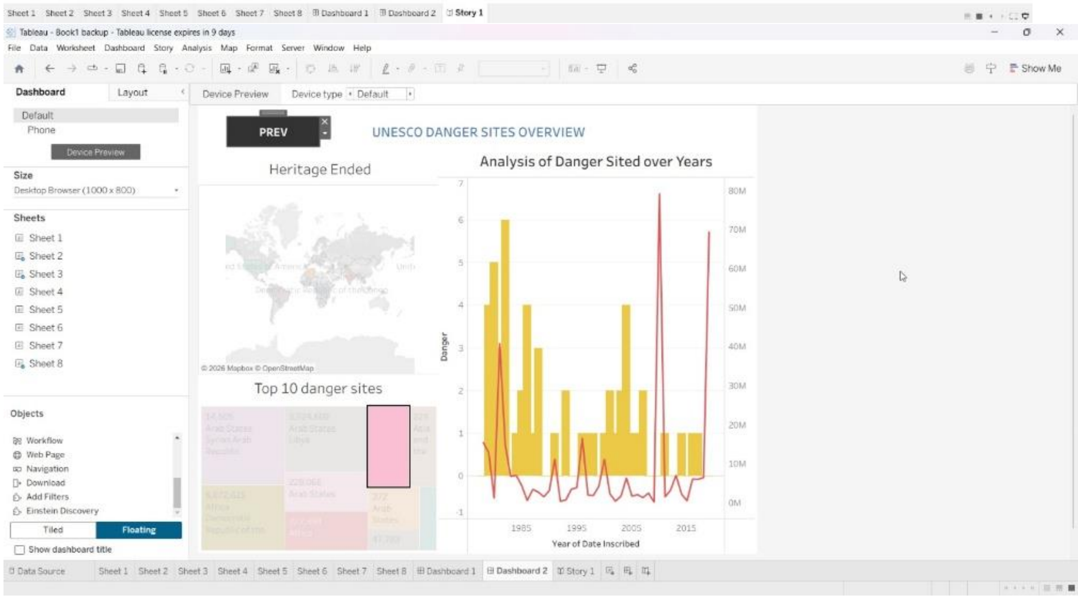


Tableau - Book1 backup - Tableau license expires in 9 days
Dashboard
Layout
Device Preview
Sheets
Objects
PREV
UNESCO DANGER SITES OVERVIEW
Heritage Ended
Analysis of Danger Sited over Years
Top 10 danger sites
Year of Date Inscribed

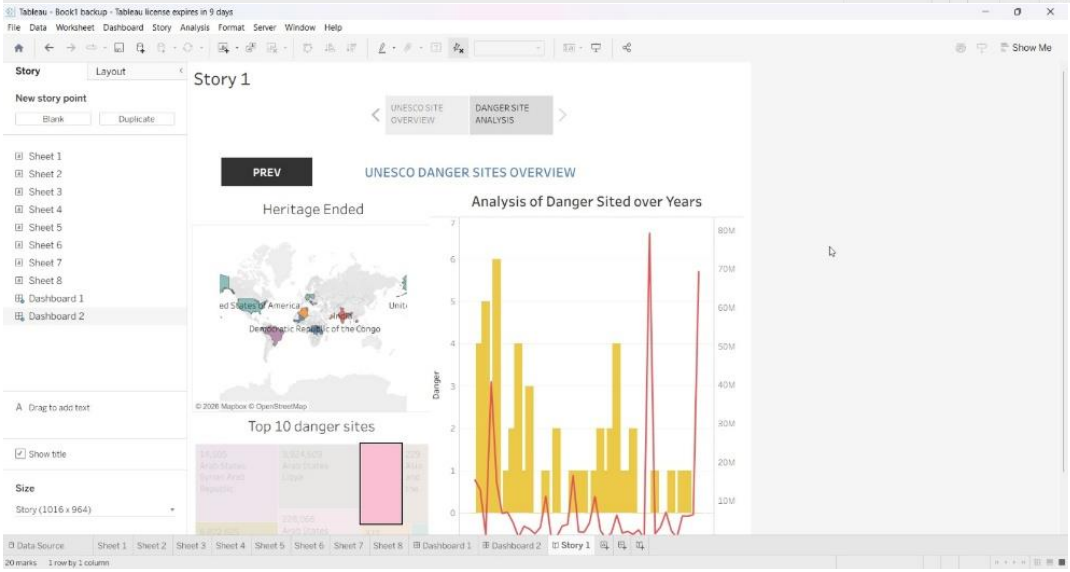


Tableau - Book1 backup - Tableau license expires in 9 days
Story
Layout
New story point
Story 1
UNESCO SITE OVERVIEW
DANGER SITE ANALYSIS
PREV
UNESCO DANGER SITES OVERVIEW
Heritage Ended
Analysis of Danger Sited over Years
Top 10 danger sites

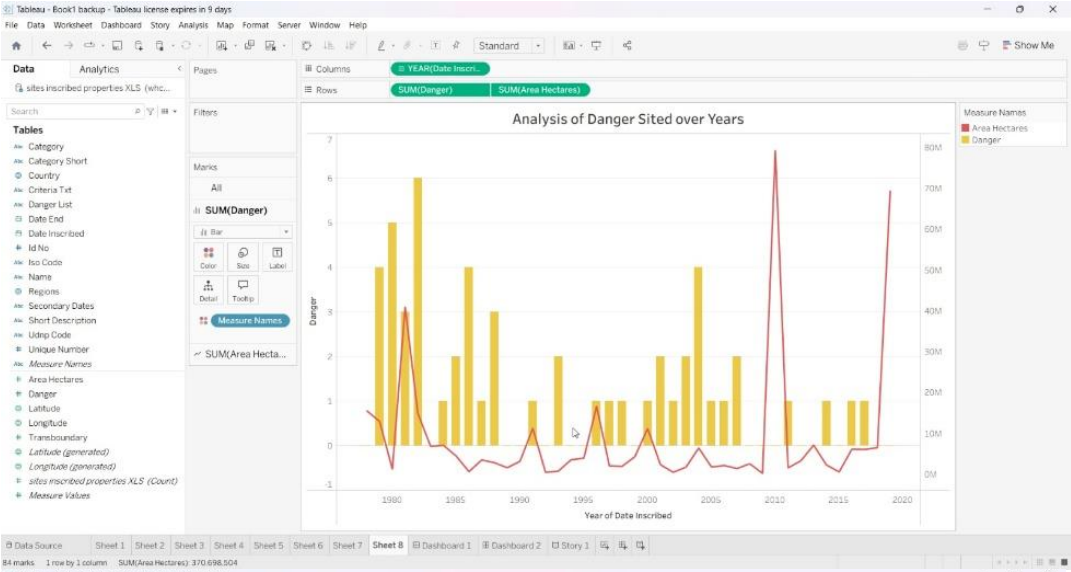


Tableau - Book1 backup - Tableau license expires in 9 days
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
Show Me
Data
Analytics
sites inscribed properties XLS (whc...
Pages
Columns
YEAR(Date Inscri...
Rows
SUM(Danger)
SUM(Area Hectares)
Filters
Marks
All
SUM(Danger)
Bar
Color
Size
Label
Detail
Tooltip
Measure Names
SUM(Area Hecta...
Tables
Category
Category Short
Country
Criteria Txt
Danger List
Date End
Date Inscribed
Id No
Iso Code
Name
Regions
Secondary Dates
Short Description
Udnp Code
Unique Number
Measure Names
Area Hectares
Danger
Latitude
Longitude
Transboundary
Latitude (generated)
Longitude (generated)
sites inscribed properties XLS (Count)
Measure Values
Analysis of Danger Sited over Years
Danger
Year of Date Inscribed
Measure Names
Area Hectares
Danger
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
Dashboard 2
Story 1
84 marks 1 row by 1 column SUM(Area Hectares): 370,698,504

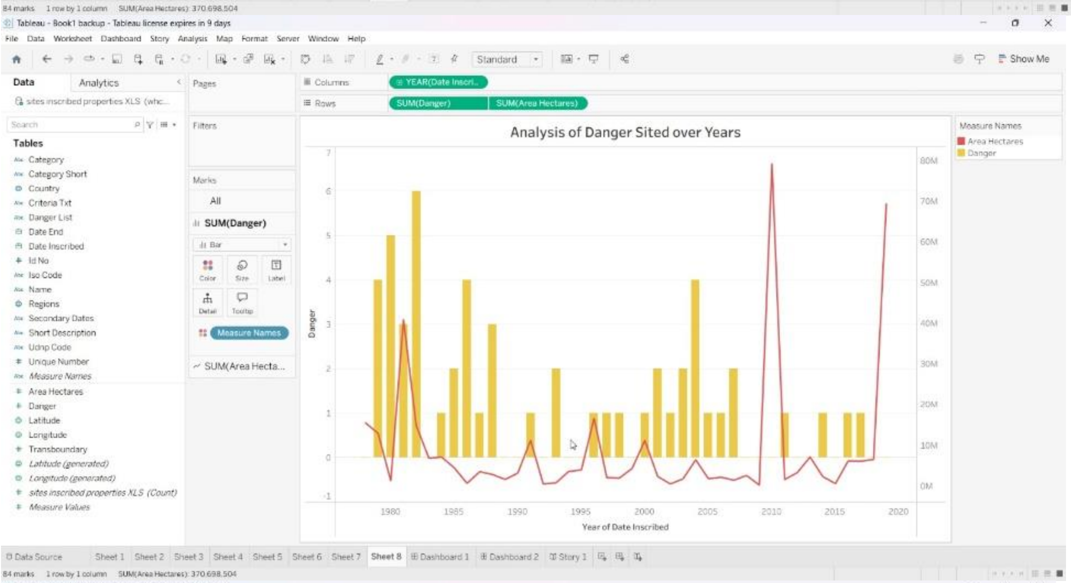



Tableau - Book1 backup - Tableau license expires in 9 days
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
Show Me
Data
Analytics
sites inscribed properties XLS (whc...
Columns
YEAR(Date Inscri...
Rows
SUM(Danger)
SUM(Area Hectares)
Analysis of Danger Sited over Years
Danger
Year of Date Inscribed
Measure Names
Area Hectares
Danger
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
Dashboard 2
Story 1
84 marks 1 row by 1 column SUM(Area Hectares): 370,698,504

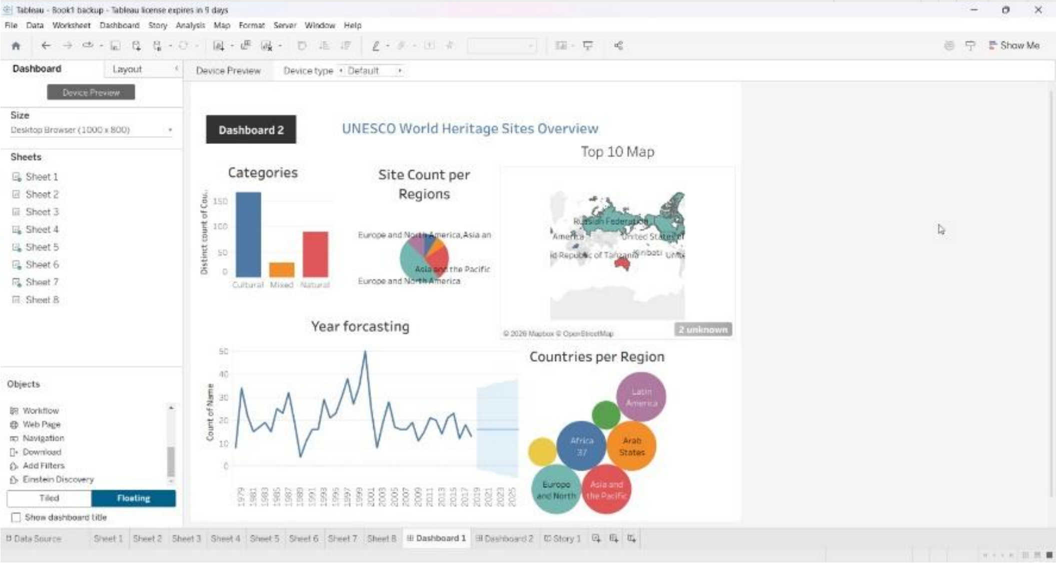



Tableau - Book1 backup - Tableau license expires in 9 days
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Show Me
Dashboard
Layout
Device Preview
Device type Default
Size
Desktop Browser (1000 x 800)
Sheets
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Objects
Workflow
Web Page
Navigation
Download
Add Filters
Einstein Discovery
Tiled
Floating
Show dashboard title
Dashboard 2
UNESCO World Heritage Sites Overview
Top 10 Map
Categories
Site Count per Regions
Cultural
Mixed
Natural
Russian Federation
United States
© 2026 Mapbox © OpenStreetMap
2 unknown
Year forcasting
Count of Name
Countries per Region
Latin America
Africa
Arab States
Europe and North
Asia and the Pacific
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
Dashboard 2
Story 1

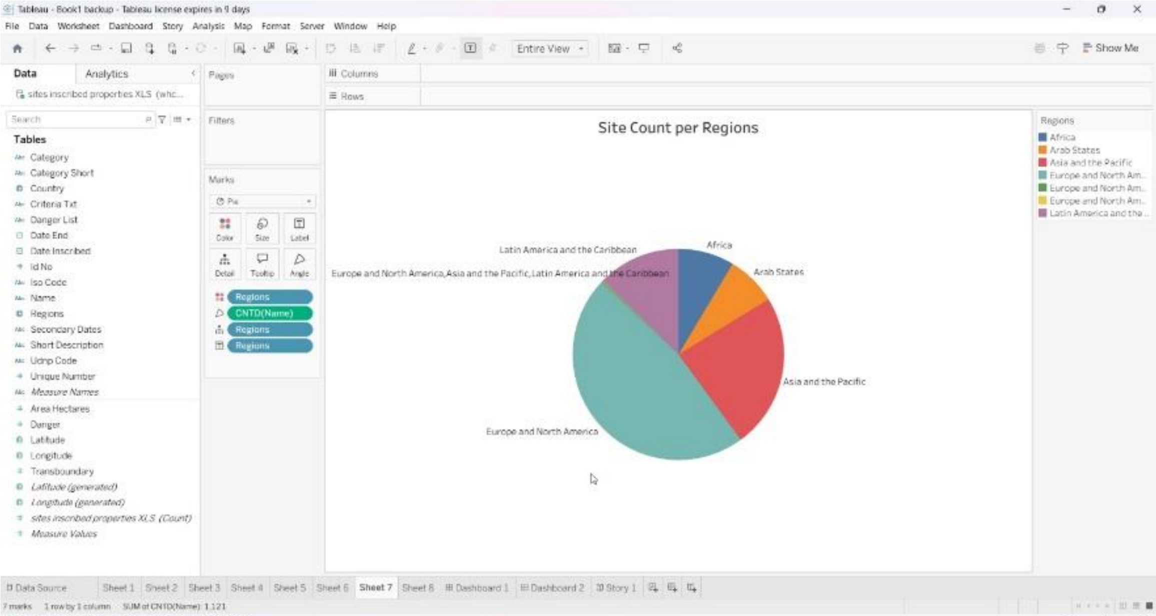


Tableau - Book1 backup - Tableau license expires in 9 days
Site Count per Regions
Latin America and the Caribbean
Africa
Arab States
Europe and North America,Asia and the Pacific,Latin America and the Caribbean
Asia and the Pacific
Europe and North America
Regions
Africa
Arab States
Asia and the Pacific
Europe and North Am..
Latin America and the..

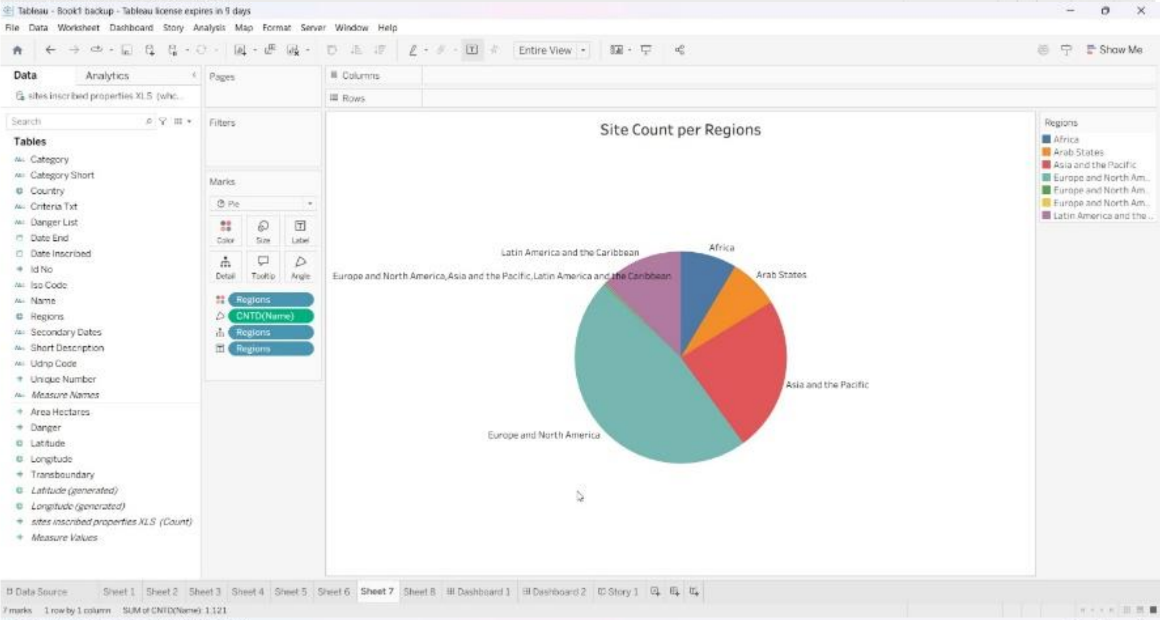


Tableau - Book1 backup - Tableau license expires in 9 days
Site Count per Regions
Latin America and the Caribbean
Africa
Arab States
Europe and North America,Asia and the Pacific,Latin America and the Caribbean
Asia and the Pacific
Europe and North America

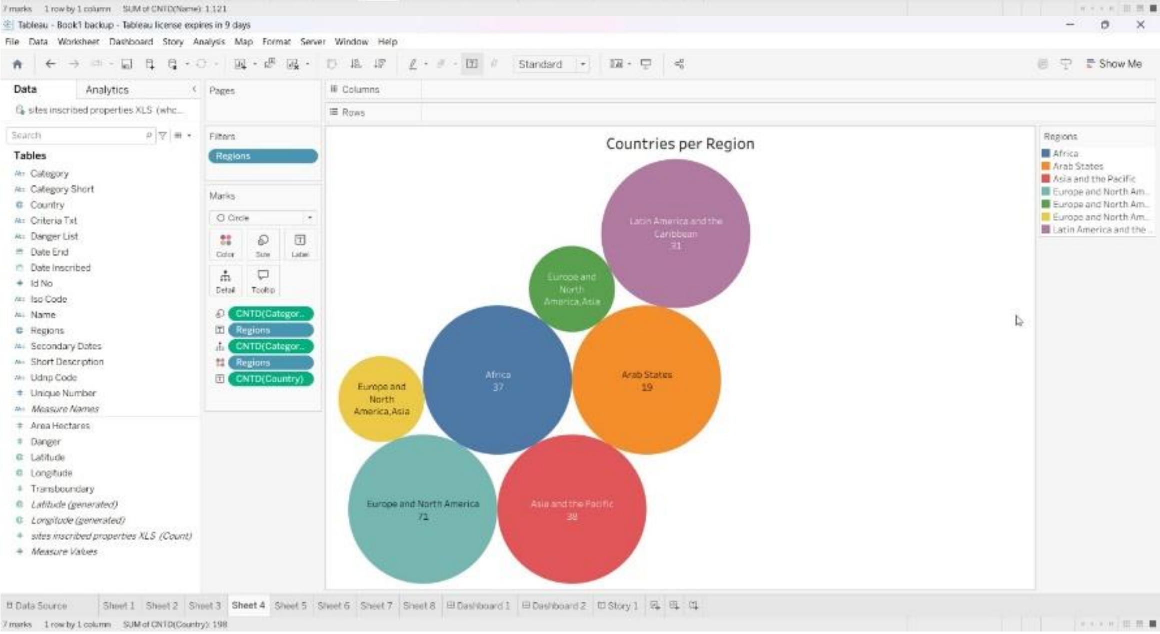


Tableau - Book1 backup - Tableau license expires in 9 days
Countries per Region
Latin America and the Caribbean 31
Europe and North America,Asia
Africa 37
Arab States 19
Europe and North America,Asia
Europe and North America 71
Asia and the Pacific 38

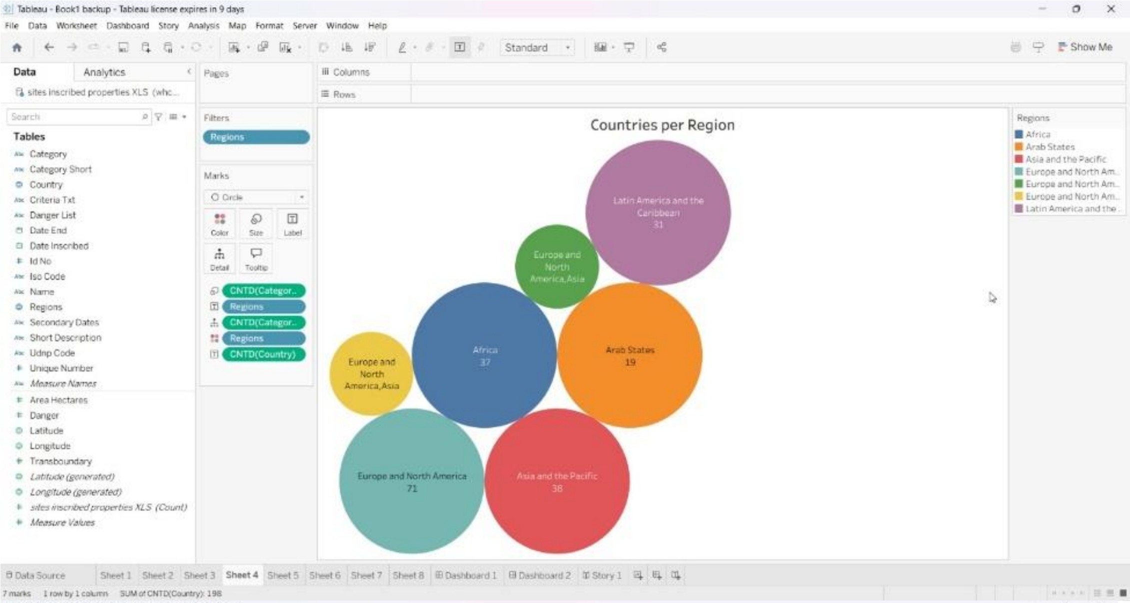


Tableau - Book1 backup - Tableau license expires in 9 days
Countries per Region
Latin America and the Caribbean
31
Europe and North America,Asia
Africa
37
Arab States
19
Europe and North America
71
Asia and the Pacific
38
Regions
Africa
Arab States
Asia and the Pacific
Sheet 4
7 marks 1 row by 1 column SUM of CNTD(Country): 198

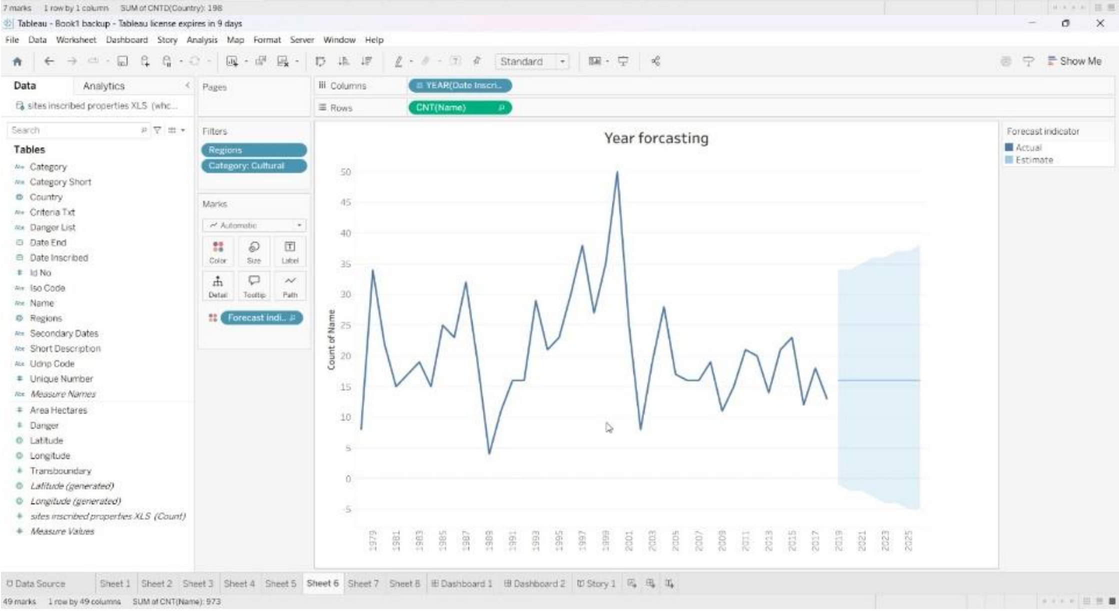


Tableau - Book1 backup - Tableau license expires in 9 days
YEAR(Date Inscri...
CNT(Name)
Regions
Category: Cultural
Year forcasting
Count of Name
Forecast indicator
Actual
Estimate
Sheet 6
49 marks 1 row by 49 columns SUM of CNT(Name): 973

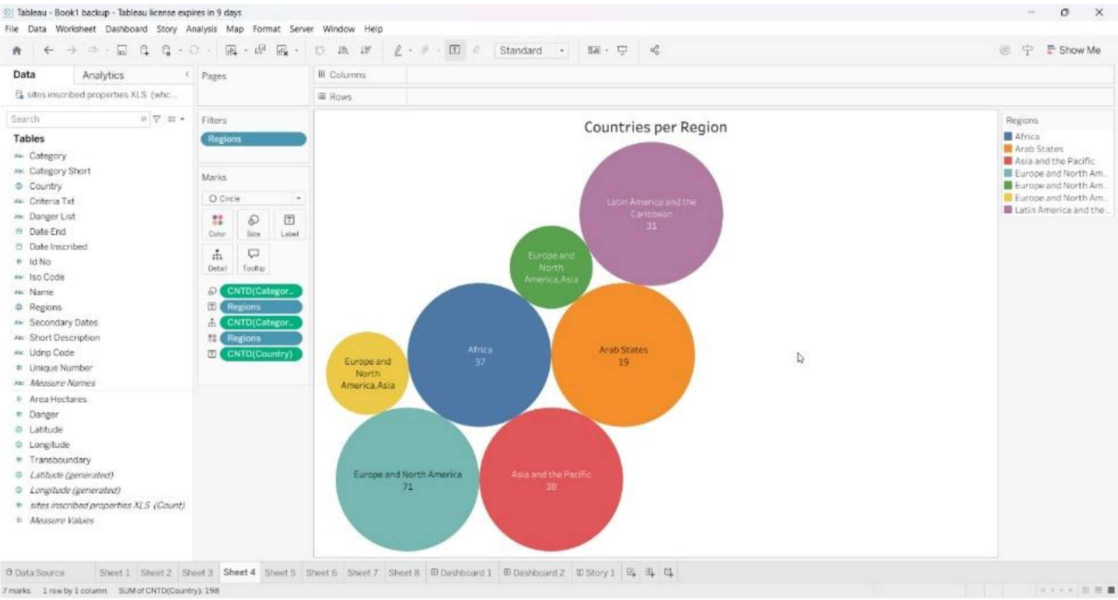


Tableau - Book1 backup - Tableau license expires in 9 days
Countries per Region
Latin America and the Caribbean
31
Europe and North America,Asia
Africa
37
Arab States
19
Europe and North America
71
Asia and the Pacific
38
Regions
Africa
Arab States
Asia and the Pacific
Sheet 4
7 marks 1 row by 1 column SUM of CNTD(Country): 198

# 8. ADVANTAGES & DISADVANTAGES:

## *8.1. Advantages:*

One of the major advantages of the UNESCO World Heritage Sites analytics project is its ability to present complex data in a clear and interactive format. By using Tableau dashboards, the system transforms raw datasets into visually appealing charts such as maps, treemaps, and trend analyses, making it easier for users to understand global heritage patterns. The integration of Tableau with a Flask web application allows users to access dashboards through a structured website, improving accessibility and user engagement.

Another important advantage is the use of modern technologies that support scalability and flexibility. Since the dashboards are hosted on Tableau Public, updates to visualizations can be reflected on the website without major code changes. The system also promotes better learning and awareness by enabling users to explore heritage site categories, regional distribution, and endangered site trends through interactive storytelling. Additionally, the lightweight Flask framework ensures simple deployment and efficient performance, making the application suitable for academic and analytical purposes.

## *8.2. Disadvantages:*

Despite its strengths, the project has certain limitations. One disadvantage is the dependency on Tableau Public for hosting dashboards, which requires an internet connection to load visualizations. If Tableau Public services are unavailable or slow, the embedded dashboards may take longer to load or may not function properly. This reliance on external hosting can affect performance in low-network environments.

Another limitation is that the application currently focuses mainly on visualization rather than advanced data processing or real-time updates. Since the dashboards use a static dataset, new data must be manually updated in Tableau before changes appear on the website. Additionally, while the Flask web interface provides a clean layout, it may require further enhancements such as advanced navigation features or responsive design improvements to support different screen sizes effectively. Addressing these limitations in future updates could make the system more robust and scalable.

# 9. CONCLUSION:

The UNESCO World Heritage Sites analytics project successfully demonstrates how data visualization and web technologies can be combined to transform complex datasets into meaningful and interactive insights. By utilizing Tableau for creating analytical dashboards and Flask for developing a web-based interface, the project provides a structured platform where users can explore heritage site distributions, category comparisons, endangered site analysis, and historical trends through visual storytelling. The integration of dashboards into a responsive website enhances accessibility and allows users to interact with data in an engaging and intuitive way.

Throughout the development process, the project followed a systematic approach that included ideation, requirement analysis, project design, implementation, and testing. Each phase contributed to building a reliable and user-friendly analytics system that highlights the importance of global heritage conservation. The final outcome not only improves understanding of UNESCO heritage data but also showcases practical skills in data analytics, dashboard development, and web integration.

In conclusion, the project achieves its goal of creating an interactive analytics platform that bridges the gap between raw data and visual insights. It serves as a valuable learning experience in applying modern tools and frameworks to real-world datasets. With future enhancements such as advanced interactivity, improved performance optimization, and expanded datasets, the system has the potential to evolve into a more comprehensive heritage analytics solution.

## 10. FUTURE SCOPE:

The UNESCO World Heritage Sites analytics project provides a strong foundation for interactive data visualization and web-based analytics, and there are several opportunities for future improvements and expansion. One possible enhancement is the integration of real-time or regularly updated datasets so that the dashboards can reflect the most recent UNESCO heritage information. Automating the data update process would reduce manual work and ensure that users always access the latest insights. Additionally, more advanced filters and interactive controls can be introduced to allow users to customize their analysis based on region, category, or time period.

Another future scope involves improving the web application by adding advanced frontend features such as responsive design for mobile devices, dynamic navigation menus, and enhanced user interface elements. The system could also be extended by incorporating additional analytical tools such as predictive analytics, machine learning models for trend forecasting, or deeper comparative analysis between countries and regions. Deploying the application on a cloud platform instead of running it locally would make the system accessible to a wider audience.

Furthermore, the project can be expanded to include educational modules or informative content that explains the significance of heritage sites and conservation efforts. Integrating user authentication features or personalized dashboards could provide a more customized experience. Overall, the future scope of this project lies in enhancing interactivity, improving performance, expanding data capabilities, and transforming the platform into a more comprehensive and scalable global heritage analytics solution.

# 11. APPENDIX:

## *11.1. Source Code:*

<u>index.html</u>

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>UNESCO World Heritage Sites
Project</title>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background-color: #f4f6f9;
      text-align: center;
    }
    header {
      background: #2c3e50;
      color: white;
      padding: 25px 10px;
    }
    header h1 {
      margin: 0;
      font-size: 32px;
    }
    .container {
      width: 90%;
      max-width: 1100px;
      margin: auto;
    }
    section {
      margin: 40px 0;
      background: white;
      padding: 20px;
      border-radius: 12px;
      box-shadow: 0 4px 12px
rgba(0,0,0,0.1);
    }
    h2 {
      color: #2c3e50;
      margin-bottom: 15px;
    }
    iframe {
      border: none;
      width: 100%;
    }
    footer {
      background: #2c3e50;
      color: white;
      padding: 15px;
      margin-top: 40px;
    }
  </style>
</head>
<body>
<header>
  <h1>UNESCO World Heritage Sites PROJECT
by LTVIP2026TMIDS89796</h1>
</header>
<div class="container">
  <!-- Dashboard 1 -->
  <section>
    <h2>Dashboard 1</h2>
    <iframe
src="https://public.tableau.com/views/Dashb
oard1-
LTVIP2026TMIDS89796/Dashboard1?:showViz
Home=no&:embed=true"
      height="820">
    </iframe>
  </section>
  <!-- Dashboard 2 -->
  <section>
    <h2>Dashboard 2</h2>
    <iframe
src="https://public.tableau.com/views/Dashb
oard2-
LTVIP2026TMIDS89796/Dashboard2?:showViz
Home=no&:embed=true"
      height="820">
    </iframe>
  </section>
  <!-- Story -->
  <section>
    <h2>Project Story</h2>
    <iframe
src="https://public.tableau.com/views/UNESC
OHERITAGEPROJECT-
LTVIP2026TMIDS89796/Story1?:showVizHom
e=no&:embed=true"
      height="950">
    </iframe>
  </section>
</div>
<footer>
  © 2026 UNESCO Heritage Tableau Project |
Built with Flask
</footer>
</body>
</html>
```

App.py

```
from flask import Flask, render_template
app = Flask(__name__)

@app.route("/")
def home():
   return render_template("index.html")
if __name__ == "__main__":
   app.run()
```

*11.2. Dataset Link:*

https://www.kaggle.com/datasets/ujwalkandi/unesco-world-heritage-sites/data?select=whc-sites-2019.csv

*11.3. Drive Link:*

https://drive.google.com/drive/folders/1JFRpCEYycBNGlCgKiaVg33OEMdVHHgU-?usp=sharing

*11.4. Demo Video Link:*

https://drive.google.com/file/d/1w4igsL8GQYRUBmvmcvD3BNOT4ep6kKN8/view?usp=sharing

*11.5. Website Link:*

*https://unescoheritageproject-ltvip2026tmids8.netlify.app/*

*11.6. GitHub Link:*

*https://github.com/venkatteja01/tableau_project*

*11.7. Tableau Public Profile:*

*https://public.tableau.com/app/profile/venkat.teja.ch/vizzes*